

Statistical and Machine Learning SS26

Exercise Sheet 2

Exercise 1: Melbourne

- a) Load the data set Melbourne_housing_FULL.csv
 - b) Subset the data to only look at 3 suburbs: Brunswick, Craigieburn and Hawthorn.
 - c) Exploratory data analysis: Get overview of dataset
 - i) Number of observations
 - ii) Number and classes of variables
 - iii) Convert them if necessary
 - iv) Remove observations with missings (only in this case)
 - v) Look more closely at variables **Price** and **Building Area**, incl. histogram/density plot
 - d) Linear regression to predict house prices
 - i) Construct a simple linear regression using only **BuildingArea** as a predictor. Interpret the coefficient. Is **BuildingArea** a significant predictor?
 - ii) Construct a multiple linear regression by adding **Suburb** as a predictor. Interpret the coefficients. Does the model improve?
 - e) Effect of outliers: Houses with very small building areas of 5sqm and lower and larger places over 300 sqm look like outliers.
 - i) Use Cook's Distance to assess whether these houses are influential in the model from d)
 - ii) What do you conclude?
 - f) Predict the price of a house in Hawthorn with 2 car spaces and 100 sqm in building area. What is the 95% prediction interval and how would you interpret it?
 - g) Suppose we categorize house price as two categories, High price ($\geq 900,000$) and low price ($< 900,000$). Construct classification models using logistic regression and linear discriminate analysis.
 - h) Compare and contrast the models in d) and g): Which model would you use to advise a buyer? What are the limitations of each approach?
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Exercise 2: Salaries

The dataset 'salary.dat' contains salary data for $n=52$ faculty members at a small US college. The variables are: **sx** (gender: female, male), **rk** (rank: assistant, associate, full), **yr** (years at rank), **dg** (highest degree: doctorate, masters), **yd** (years since degree), and **s1** (salary, the response variable).

- a) Linear regression
 - Regress **s1** on **sx** and interpret the resulting coefficients. Is there evidence of gender discrimination in salaries?
 - b) Controlling for differences
 - i) Why could the estimate of the gender effect in a) be biased?
 - ii) Fit a multiple linear regression of **s1** on **sx**, **dg**, **yd**, and the interaction **dg:yd**. Interpret all coefficients
 - iii) Does controlling for qualification and experience change the conclusion about gender discrimination?
 - c) Jackknife residuals
 - i) Compute the jackknife residuals (manually or with R function) and plot them against the fitted values
 - ii) What does the plot indicate?
 - iii) Comment on any observations with jackknife residuals exceeding 2 in absolute value. Can these formally be considered outliers?
 - d) Leverage and Cook's distance
 - i) Compute the leverage h_{ii} and Cook's distance D_ℓ (manually or with R function) for all observations
 - ii) List any high-leverage points (exceeding $2p/n$). Does high leverage translate into actual influence?
 - iii) Identify the most influential observation and discuss whether removing it would change your conclusions from b)
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Exercise 3: Linear models

Linear Regression

The **Boston** dataset contains housing data for 506 census tracts in Boston. To find out more about the data set: type `?Boston`. We aim to predict **medv** (median house value) using **lstat** (percent of households with low socioeconomic status).

- Load the data (The **ISLR2** library contains this data set. Installation of **ISLR2** package via `install.packages("ISLR2")`). Get an overview of the data.
- Fit a simple linear regression with **medv** as the response and **lstat** as the predictor. Interpret the estimated coefficients. Are they statistically significant?
- Plot **medv** against **lstat** and add the fitted regression line. Does a linear relationship seem appropriate?
- Plot the residuals against the fitted values. What do you observe? Which model assumption might be violated?
- Compute a 95% confidence interval and a 95% prediction interval for **medv** at **lstat** = 5, 10, 15. What is the difference between the two intervals? Why is the prediction interval wider?
- Perform a multiple linear regression using all 12 predictors - does it improve the prediction?
- Why is using a classification model to predict **medv** not appropriate here? What would need to change in order to apply logistic regression or LDA?

Logistic Regression

We will use the **Smarket** data, which is part of the **ISLR2** library. This data set consists of percentage returns for the S&P 500 stock index over 1250 days, from the beginning of 2001 until the end of 2005. For each date, there are the percentage returns for each of the five previous trading days, **Lag1** through **Lag5**, **Volume** (the number of shares traded on the previous day, in billions), **Today** (the percentage return on the date in question) and **Direction** (whether the market was Up or Down on this date). Our goal is to predict **Direction** (a qualitative response) using the other features.

- Load the **Smarket** data, get an overview of the data
- Why is linear regression not appropriate to predict **Direction**? What makes logistic regression a better choice here?
- Fit a logistic regression of **Direction** on **Lag1** through **Lag5** and **Volume**. Interpret the coefficients and their statistical significance. The `glm()` function can be used to fit many types of generalized linear models, including logistic regression.

Linear Discriminant Analysis

- Perform LDA predicting **Direction** from **Lag1** and **Lag2** on the **Smarket** data using the `lda()` function, which is part of the **MASS** library.
- Interpret the prior probabilities and group means. What do they tell you about the data?